

Introduction to HTML

A web application consists of two main components:

- Front-end (Client-Side): This is what the user interacts with. It consists of HTML, CSS, and JavaScript. Runs in the user's web browser.
- Back-end (Server-Side) : Handles business logic, database operations, and user authentication. Uses server-side languages (e.g., Python, PHP, Node.js, Java). Interacts with databases (e.g., MySQL, MongoDB).

In this course, we will focus on the front-end of web applications, which includes:

HTML (HyperText Markup Language) → Structur

CSS (Cascading Style Sheets) → CSS is used to style and format the HTML elements.

JavaScript → JavaScript adds interactivity to a web page. It allows us to create dynamic content, animations, and real-time updates.

What is HTML? (HyperText Markup Language)

HTML was first introduced by Tim Berners-Lee in 1991 as a simple way to structure documents and link them together using hyperlinks.

Over the years, HTML has evolved through different versions, from HTML 2.0 to the current HTML5, introducing new elements and features to enhance web development.

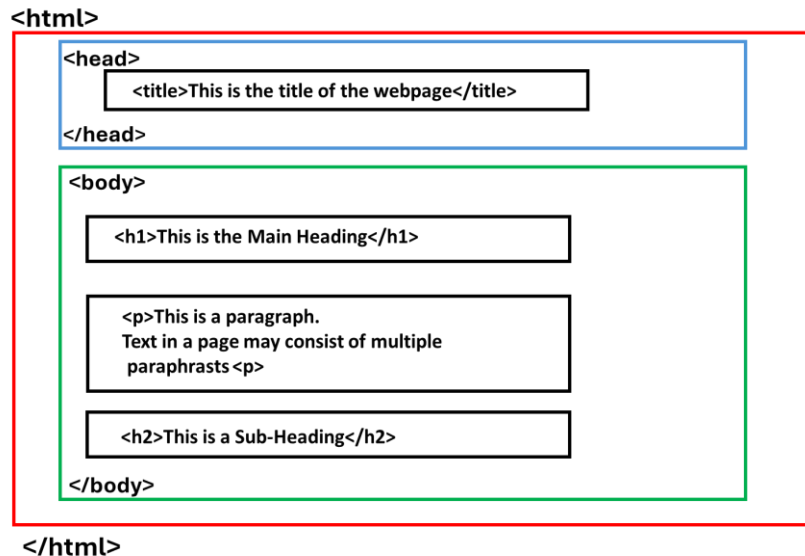
HTML is the foundation of any web page. It defines the structure of a webpage using tags.

Key Features of HTML Uses tags (e.g., <h1>, <p>, <a>,). Defines the content of a web page. HTML does not need a compiler (just an interpreter)

Example: Notepad + Internet Explorer (or other browsers like Google Chrome, Firefox, etc.)

HTML Structure

The basic structure of HTML page looks like this:



- **<html>**: The opening **<html>** tag indicates that anything between it and a closing **</html>** tag is HTML code.
- **<head>**: This tag represents the document's header which can keep other HTML tags like **<title>**, **<link>** etc.
- **<title>**: The **<title>** tag is used inside the **<head>** tag to mention the document title.
- **<body>**: tag indicates that anything between it and the closing **</body>** tag should be shown inside the main browser window.

Tags in HTML:

In HTML, there are opening tags and closing tags. The characters in the brackets indicate the tag's purpose. For example, in the tags above the **p** stands for paragraph. The closing tag has a forward slash after the **<** symbol.

<p>
Opening Tag

</p>
Closing Tag

An **Element** comprises the **opening tag** and the **closing tag** and any content that lies between them.

<p>Text</p>
Element

Similarly, an **Attribute** provide additional information about the contents of an element. They appear on the opening tag of the element and are made up of two parts: a name and a value, separated by an equals sign.

`<p style="text-align: center;">Text</p>`
Attribute

Basic Tags in HTML

Heading Tags: Any document starts with a heading. You can use different sizes for your headings. HTML also has six levels of headings, which use the elements `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`. While displaying any heading, the browser adds one line before and one line after that heading.

<code><h1> Hello</h1></code>	Hello
<code><h2> Hello</h2></code>	Hello
<code><h3> Hello</h3></code>	Hello
<code><h4> Hello</h4></code>	Hello
<code><h5> Hello</h5></code>	Hello
<code><h6> Hello</h6></code>	Hello

Paragraph Tag `<p>`: The `<p>` tag offers a way to structure your text into different paragraphs. Each paragraph of text should go in between an opening `<p>` and a closing `</p>` tag.

<code><p>This is a paragraph.</p></code>	This is a paragraph.
--	----------------------

Line Break Tag `<br>`: Whenever you use the `<br>` element, anything following it starts from the next line. This tag is an example of an empty element, where you do not need opening and closing tags, as there is nothing to go in between them.

<code><p>This is a paragraph without break line.</p></code>
<code><p> this is a paragraph
 with break line tag</p></code>

Output:

This is a paragraph without break line.

this is a paragraph
with break line tag

Horizontal lines <hr>: are used to visually break-up sections of a document. The <hr> tag creates a line from the current position in the document to the right margin and breaks the line accordingly.

```
<p>This is a paragraph.</p>
<hr>
<p> This is another paragraph</p>
```

Output:

This is a paragraph.

This is another paragraph

Bold : Anything that appears within ... element, is displayed in bold

Strong : Anything that appears within ... element is displayed as important text.

Italic <i>: Anything that appears within <i>...</i> element is displayed in italicized

Emphasis : Anything that appears within this element is displayed as emphasized in italic.

```
<b>This text is bold.</b>
<strong>This text is important.</strong>
<i>This text is italicized.</i>
<em>This text is emphasized.</em>
```

Output:

This text is bold. This text is important. *This text is italicized. This text is emphasized.*

Comment Tag: comments can be added to HTML source by using the following syntax: <!--
Write comments here -->

Superscript <sup>: The content of a ^{...} element is written in superscript; the font size used is the same size as the characters surrounding it but is displayed half a character's height above the other characters.

Subscript <sub>: The content of a _{...} element is written in subscript; the font size used is the same as the characters surrounding it, but is displayed half a character's height beneath the other characters.

```
H<sup>2</sup>O  <!-- Superscript example -->  
CO<sub>2</sub>  <!-- Subscript example -->
```

H²O
CO₂

Highlight <mark>: Anything that appears with-in <mark>...</mark> element, is displayed as marked with yellow ink.

```
<mark>This text is highlighted.</mark>
```

Output:

This text is highlighted.

Image : The tag is used to embed images in a webpage. The src attribute specifies the path to the image file. Other attributes control aspects such as alternative text, width, height.

```

```

Output:



HTML Grouping Using <div> and

- **<div> (Division Tag):** A block-level element used to group multiple elements together for styling or layout purposes.
- ** (Span Tag):** An inline element used to apply styles or scripts to a specific portion of text within a block.

HTML Lists

Lists help organize content in a structured manner. They enhance readability and improve the presentation of information. Lists are widely used in navigation menus, feature lists, instructions, and more.

Types of HTML Lists

- **Ordered Lists ():** Used when the order of items is important.
- **Unordered Lists ():** Used when order does not matter.
- **Definition Lists (<dl>):** Used for presenting terms and their descriptions.

Ordered Lists ()

- Ordered lists are used when the sequence of items is important.
- Each item is numbered or labeled sequentially (e.g., 1, 2, 3 or I, II, III).
- An ordered list starts with the tag. Each list item starts with the tag.
- The list items will be marked with numbers by default

```
<!-- Ordered List (ol) -->  
<ol>  
  <li>First item</li>  
  <li>Second item</li>  
  <li>Third item</li>  
</ol>
```

Output:

1. First item
2. Second item
3. Third item

Key Attributes of Ordered Lists

- **type:** Defines the numbering style (e.g., "1", "A", "a", "I", "i").
- **start:** Specifies the starting value of the numbering.
- **reversed:** Displays numbers in descending order.

```
<!-- Ordered List (ol) -->
<ol type="A">
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ol>
```

A. First item
B. Second item
C. Third item

Unordered Lists ()

- Unordered lists are used when the order of items does not matter.
- An unordered list starts with the tag. Each list item starts with the tag.
- Items are displayed with bullet points instead of numbers.
- Different styles of bullets can be applied using CSS.

```
<!-- Unordered List (ul) -->
<ul>
  <li>Item one</li>
  <li>Item two</li>
  <li>Item three</li>
</ul>
```

Output:

- Item one
- Item two
- Item three

Definition Lists (<dl>)

- Definition lists are used for presenting terms and their corresponding descriptions.
- They consist of:
 - <dl> (Definition Lists): tag defines the description list
 - <dt> (Definition Term): Represents the term being defined.
 - <dd> (Definition Description): Contains the explanation or description of the term.
 -

```
<!-- Definition List (dl) -->
<dl>
  <dt>HTML</dt>
  <dd>A markup language for creating web pages.</dd>

  <dt>CSS</dt>
  <dd>A style sheet language used for describing the look of a document.</dd>
</dl>
```

Output:

HTML
A markup language for creating web pages.

CSS
A style sheet language used for describing the look of a document.

Nesting Lists

- Lists can be nested by placing one list inside another.
- This is commonly used for multi-level navigation menus or hierarchical content.

Code Sample

```
<html>
  <head>
    <title>HTML Examples</title>
  </head>
  <body>
    <!-- Headings -->
    <h1>Heading 1</h1>
    <h2>Heading 2</h2>
    <h3>Heading 3</h3>
    <h4>Heading 4</h4>
    <h5>Heading 5</h5>
    <h6>Heading 6</h6>
    <!-- Paragraph and Line Break -->
    <p>This is a paragraph of text.</p>
    <p>This is another paragraph.<br>This text appears on a new
line.</p>
    <!-- Horizontal Line -->
    <hr>
    <!-- Text Formatting -->
    <b>Bold Text</b><br>
    <strong>Strong (Important) Text</strong><br>
```



```
<i>Italicized Text</i><br>
<em>Emphasized Text</em><br>
<mark>Highlighted Text</mark><br>
<sup>Superscript Example (e.g., x2)</sup><br>
<sub>Subscript Example (e.g., H2O)</sub><br>
<!-- Comment -->
<!-- This is a comment in HTML and will not be displayed in the
browser -->
<!-- Image -->
<h3>Image Example</h3>


<!-- Grouping Elements -->
<div style="background-color: lightgray;">
  <h3>Div Example</h3>
  <p>This is inside a div container.</p>
</div>
<p>This is an example of <span style="color: red;">inline
text</span> using the <code>&lt;span&gt;</code> tag.</p>

<!-- Lists -->
<h3>Ordered List</h3>
<ol>
  <li>Item One</li>
  <li>Item Two</li>
  <li>Item Three</li>
</ol>
<h3>Unordered List</h3>
<ul>
  <li>Bullet Item 1</li>
  <li>Bullet Item 2</li>
  <li>Bullet Item 3</li>
</ul>
<h3>Definition List</h3>
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
```

```
    </dl>  
</body>  
</html>
```